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Method and apparatus for producing face protection masks

Abstract

A method and an apparatus for manufacturing, in a continuous flow, face protection masks including a mask body and two U-shaped elastic thread portions, which are formed by bending respective elastic thread portions retained by a group of sectors including a fixed sector and two movable sectors rotatable with respect to the fixed sector.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 17/540,422 filed on Dec. 2, 2021, and claims priority to Italian Patent Application No. 102020000029588 filed on Dec. 3, 2020, the entire disclosure of these applications is incorporated herein by reference.

FIELD OF THE INVENTION
(1) The present invention relates to a method and an apparatus for producing face protection masks.
(2) More specifically, the invention relates to a method and an apparatus for producing face protection masks comprising a mask body and a pair of elastic ear loops attached to opposite sides

of the mask body.
(3) The invention may be employed both for the production of disposable surgical masks and for the production of FFP (Filtering Face Piece) filtering masks having a FFP1, FFP2 or FFP3 protection level, according to the standard DIN EN 149:2001+A1:2009.

DESCRIPTION OF THE RELATED ART

(4) Disposable surgical masks and filtering masks typically comprise a mask body and two C-shaped elastic threads attached to opposite sides of the mask body, which thereby form two ear loops to be arranged around the ears of the user in order to keep the mask body in position against the user's mouth and nose.
(5) Most known apparatuses for producing face protection masks of this type envisage stopping the mask bodies in order to apply the elastic ear loops. The ends of the elastic thread portions forming the ear loops are gripped by pliers and applied to the respective mask body. Then, the ends of the elastic thread portions are welded to respective portions of the mask body while the latter is kept stationary. Such apparatuses have a typical production speed of about 50-80 pieces/1′. In order to increase the production speed of such machines, the mask bodies are normally shunted towards two or more welding units, whereat the elastic ear loops are attached to the mask bodies.
(6) In order to increase the speed of the apparatuses for producing face protection masks, solutions have already been proposed wherein the elastic threads which will form the ear loops are bent into a sinusoidal shape before being attached to the mask bodies. Such solutions enable increasing the production speed of the apparatus, but they suffer from problems related to an insufficient control of the tension and of the length of the elastic threads. Indeed, during the sinusoidal bending of the elastic threads, it is very difficult to ensure an accurate control of the thread tension. As elastic threads are extensible, a variation in tension involves a variation in length. Therefore, a problem of such known solutions consists in the possibility that the elastic thread portions forming the elastic ear loops may not have uniform lengths. A further problem in known apparatuses is that the bending of the elastic threads into a sinusoidal shape causes the ends of the elastic threads not be orthogonal but rather inclined with respect to the mask body side edges, at the expense of the ergonomic properties of the product and with a consequent greater length of the thread portions forming the welding areas.

OBJECT AND SUMMARY OF THE INVENTION

(7) The present invention aims at providing a method and an apparatus for producing face protection masks which overcome the problems of the related art.
(8) Specifically, the present invention aims at providing a method for producing face protection masks having a production speed higher than the known apparatuses.
(9) A further object of the present invention is providing a method and an apparatus for producing face protection masks which enable obtaining a higher uniformity of the length of the elastic thread forming the ear loops.
(10) According to the present invention, said objects are achieved through a method and an apparatus that has the features set forth in claim **1**.
(11) The Claims form an integral part of the teaching provided herein with reference to the invention.
(12) According to the present invention, forming the ear loops is obtained by retaining successive portions of elastic threads through arrays of holding sectors being arranged in subsequent groups, wherein in each group a central sector is fixed and two lateral sectors are movable, in order to bend the respective portion of elastic thread into a U-shape.
(13) Thanks to this inventive solution it is possible to bend the elastic threads into a U-shape at high speed and without ever losing the control on the thread tension. Therefore, the U-bending is performed while keeping control of the length of the elastic thread portions.
(14) In possible embodiments, the U-bent ends of the elastic thread portions may be fixed to a continuous tape formed by at least one continuous web of non-woven fabric. The fixation between

the ends of the elastic thread portions and the tape may be obtained via gluing, ultra-sonic welding, thermal welding or a mechanical anchoring of the threads between the welding spots. The tape which keeps the elastic thread portions bent in a U-shape may be fixed to the respective mask body via gluing, ultra-sonic welding or thermal welding.
(15) In possible embodiments, the ends of the U-bent elastic thread portions may be attached directly to the respective mask bodies, without being previously fixed to a tape.
(16) When the ends of the U-bent elastic threads are fixed to a tape, said tape may be continuous and may be cut in a longitudinal direction in order to form two parallel arrays of ear loops.
(17) In possible embodiments, the tape which retains the U-bent elastic thread portions may be formed by discrete portions of non-woven fabric.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present invention will now be described in detail with reference to the annexed drawings, which are given by non-limiting example only, wherein:
(2) FIG. **1** is a perspective view of an embodiment of a face protection mask produced through a method and an apparatus according to the present invention,
(3) FIG. **2** is an exploded perspective view of the face protection mask of FIG. **1**,
(4) FIG. **3** is a schematic view of an apparatus for producing the face protection masks of FIGS. **1** and **2**,
(5) FIG. **4** is a schematic perspective view showing a part of the apparatus of FIG. **3**,
(6) FIG. **5** is a schematic plan view showing the process steps for forming the elastic ear loops, and
(7) FIG. **6** is a schematic plan view showing an embodiment of a component of the apparatus of FIG. **2**.
(8) It will be appreciated that the various Figures may not be drawn to the same scale. Moreover, it will be appreciated that in some Figures certain elements or components may not be shown, in order to better highlight other elements or components or in order to simplify the comprehension of the Figures.
DETAILED DESCRIPTION
(9) With reference to FIGS. **1** and **2**, reference **10** denotes a face protection mask according to an embodiment of the present invention. Mask **10** comprises a mask body **12** and two ear loops **14**.
(10) The mask body **12** is made of a non-woven fabric and has a flat rectangular shape, with two longer sides **16** parallel to a longitudinal axis A and two shorter sides **18** orthogonal to longitudinal axis A. In a possible embodiment, the mask body **12** may be formed by three layers of non-woven fabric overlapping one another. For example, the inner layer may be a Thermal-Bond material, the intermediate layer a Meltblown material and the outer layer a Spunbond material.
(11) The mask body **12** has a series of pleats **20** which are parallel to the longitudinal axis A, and which extend throughout the length of the mask body **12**. The mask body **12** has an outer face **22** and an inner face **24** parallel to each other. The mask body **12** has respective lateral portions adjacent to the respective shorter sides **18**.
(12) Each of both ear loops **14** includes an elastic thread portion **28** which is bent into a U-shape, and a portion of non-woven tape **30**. The elastic thread portion **28** of each ear loop **14** may be made of Spandex. The elastic thread portion **28** has opposite ends **28′** fixed to the tape portion **30**. The ends **28′** of the elastic thread portion **28** may be fixed to the tape portion **30** via gluing, ultra-sonic welding, thermal welding or a mechanical anchoring between two welding spots (as described in EP-B1-0886480 by Cera France). The tape portion **30** may be formed either by a single layer of non-woven fabric or by two layers of non-woven fabric, which are attached to each other and arranged at opposite sides of the ends **28′** of the elastic thread portion **28**. The layer(s) of non-

woven fabric forming the tape portions **30** may be formed of a Thermal-Bond material.
(13) The tape portions **30** of both ear loops **14** may be attached to the inner face **22** or to the outer face **24** of the mask body **12**. The tape portions **30** overlap respective lateral portions **26** of the mask body **12**, and are fixed frontally to the inner face **24** or to the outer face **22** via ultra-sonic welding, gluing or thermal welding. The elastic thread portions **28** may face inwardly of the mask body **12**, as shown in FIG. **1**, or else they may extend outwardly of the mask body **12**, beyond the shorter sides **18**.
(14) The mask body **12** may include a nose clip **34**, formed by a strip of plastically deformable material which is arranged parallel to one of the longer sides **16**, with the purpose of shaping the upper edge of the mask body **12** around the user's nose. The nose clip **34** may for example be formed by a thin metal strip coated with plastic material, e.g. polyethylene. The nose clip **34** may be retained within a folded longitudinal edge **36** of the mask body **12**. The folded longitudinal edge **36** may be fixed to the outer face **22** or to the inner face **24** via ultra-sonic welding, gluing or thermal welding. In a possible embodiment, both longitudinal edges **36** may be folded and fixed to the outer face or to the inner face **24** via ultra-sonic welding, gluing or thermal welding. In possible embodiments, the nose clip **34** may be arranged between the mutually overlapping non-woven layers which form the mask body **12**.
(15) With reference to FIG. **3**, an apparatus and a method for producing a face protection mask **10** of this type will be described.
(16) In FIG. **2**, reference **40** denotes a continuous flow apparatus for producing face protection masks **10**. The phrase “continuous flow apparatus” denotes an apparatus wherein the blanks are continuously moved, without stops or interruptions, until the finished product is obtained.
(17) Apparatus **40** comprises a mask body forming unit **42**, wherein a continuous web of non-woven fabric **44** advances in the machine direction X. The forming unit **42** may be structured as described in detail in the Italian Patent Application number 10 2020 00000 8861 of the same Applicant (not yet published at the filing date of the present Application).
(18) The mask body forming unit **42** comprises a folding device **48**, configured to form, onto the continuous web of non-woven fabric **44**, continuous pleats parallel to the machine direction X. The mask body forming unit **42** comprises a nose clip application device (not shown herein) which applies, to the continuous web of non-woven fabric **44**, a sequence of nose clips, which are mutually spaced in the machine direction.
(19) Apparatus **40** comprises a cutting unit **64**, configured to cut the continuous web of non-woven fabric **44** in a direction transverse to the machine direction X, in order to form mask bodies having respective longitudinal axes parallel to the machine direction X.
(20) Downstream cutting unit **64**, the mask bodies are fed to a rotating device **66**, which is configured to individually rotate the mask bodies by 90°. Directly downstream rotating device **66**, the mask bodies have the longitudinal axes thereof arranged transversally with respect to the machine direction X. Directly downstream rotating device **66** the mask bodies **12**, oriented with their respective longitudinal axes orthogonal to the machine direction X, are picked up by a transfer roller **67**.
(21) Apparatus **40** comprises an ear loop forming unit **68**, configured to form in a continuous flow a first and a second array of ear loops.
(22) The ear loop forming unit **68** comprises two unwinders **70** (only one of which is visible in FIG. **3**) feeding two continuous elastic threads **72** parallel to each other from respective coils **74**.
(23) The ear loop forming unit **68** comprises a movable structure **76**, to which the two continuous elastic threads **72** are fed. In the embodiment shown in the Figures, the movable structure includes a wheel **76** rotatable around an axis B orthogonal to the feeding direction of the continuous elastic threads **72**. In possible embodiments, the movable structure **76** may consist in a belt or a table, which translate along a straight direction. In the following part of the description specific reference will be made to a movable structure including a wheel, but it is understood that the invention is not

limited to this specific geometry.
(24) Both continuous elastic threads **72** are feed onto the outer surface of wheel **76** along two circular arcs which are parallel to each other.
(25) The ear loop forming unit **68** includes two unwinders **78**, **80**, configured to feed onto the outer surface of wheel **76** a first web of non-woven fabric **82** and a second web of non-woven fabric **84**. Unwinders **78**, **80** may be associated to respective glue dispensers **86**, **88**, which intermittently apply layers of glue onto the upper surfaces of the respective non-woven webs **82**, **84**.
(26) In a possible embodiment one single unwinder, **78** or **80**, may be provided, which feeds onto the outer surface of wheel **76** one single non-woven web **82** or **84**.
(27) FIG. **5** shows a flat representation of the external surface of wheel **76**. Referring to FIGS. **4** and **5**, the surface of wheel **76** is provided with two holding areas, each of which is configured to retain a portion of a respective elastic thread **72**. The elastic threads **72** retained on the outer surface of wheel **76** move together with the wheel **76** in the direction shown by arrow C.
(28) Each of both holding areas includes a plurality of fixed sectors **90** and a plurality of movable sectors **92**. The fixed sectors **90** and the movable sectors **92** are provided with respective holding means, adapted to retain a respective elastic thread portion **72**. The holding means of sectors **90**, **92** may be formed by holes connected to a suction source.
(29) The fixed sectors **90** are fixed with respect to wheel **76** and are parallel to the movement direction C. The movable sectors **92** are rotatable with respect to wheel **76** around respective radial axes. Each of the movable sectors **92** may move between a first position, wherein it is parallel to the movement direction C, and a second position wherein it is transverse with respect to the movement direction C. The movable sectors **92** may be controlled by a cam and, while wheel **76** is being rotated, they move cyclically from the first position to the second position, and vice-versa.
(30) The fixed sectors **90** and the movable sectors **92** are arranged in successive groups of sectors **94**. Each group of sectors **94** comprises a fixed sector **90** and two movable sectors **92**. In each group of sectors **94**, the two movable sectors **94** are arranged laterally on opposite sides of the respective fixed sector **90**.
(31) Wheel **76** includes a plurality of anvil elements **96**, each of which is arranged between two groups of sectors **94** adjacent to each other.
(32) With reference to FIG. **4**, the ear loop forming unit **68** includes a cutting roller **98** having blades **100** cooperating with the anvil elements **96** of wheel **76**, to cut the elastic threads **72** into successive portions while they are being retained on the outer surface of wheel **76** by the respective groups of sectors **94**.
(33) With reference to FIGS. **4** and **5**, in operation both continuous elastic threads **72** are fed to an entry section of wheel **76**, wherein the elastic threads **72** are aligned with respective arrays of holding sectors **90**, **92**. On the outer surface of wheel **76**, the elastic threads **72** are retained by the respective fixed sectors **90** and movable sectors **92**. Upstream cutting roller **98**, the movable sectors **92** are aligned with the fixed sectors **90**.
(34) The elastic threads **72** are fed onto the outer surface of wheel **76** with a very small longitudinal extension. The longitudinal extension of the elastic threads **72** (i.e. the variation percentage of the length with respect to the length of the thread in an undeformed condition) may be lower than 30% and preferably comprised in a range between 10% and 30%.
(35) The first non-woven web **82** is fed onto the outer surface of wheel **76** upstream cutting roller **98**, and it is arranged on the outer surface of the wheel **76** between both elastic threads **72**. On the upper surface of the first non-woven web **82** there are intermittently applied layers of glue which are mutually spaced in the longitudinal direction.
(36) The cutting roller **98** cuts the elastic threads at regular intervals on the anvil elements **96**, between each pair of groups of sectors **94**, thereby forming a succession of elastic thread portions **28**.
(37) Downstream cutting roller **98**, each elastic thread portion **28** is retained on wheel **76** by a fixed

sector **90** and by two movable sectors **92**. After cutting the elastic threads **72**, the movable sectors **92** of each group of sectors **94** rotate by 90° towards each other, and U-bend the respective elastic thread portion **28** as shown in FIG. **5**. During U-bending, the elastic thread portions **28** are retained by the respective sectors **90**, **92**, and this ensures a uniform length of the various elastic thread portions **28**.
(38) After U-bending the elastic thread portions **28**, the ends **28′** of the elastic thread portions **28** are placed on the first non-woven web **82** at the positions of the layers of glue which have previously been applied on the first non-woven web **82**, in a precise phase relationship with respect to the wheel **76**.
(39) Referring to FIGS. **4** and **5**, after U-bending the elastic thread portions **28**, the second non-woven web **84** is applied onto the surface of wheel **76** in a position overlapping the first non-woven web **82**. Also on the second non-woven web **84** it is possible to apply intermittent layers of glue in phase relationship with wheel **76**, so that the layers of glue of the second non-woven web **84**, in the same way as the layers of glue of the first non-woven web **82**, are aligned with the respective ends **28′** of the elastic thread portions **28**.
(40) With reference to FIGS. **4** and **5**, at the exit of wheel **76**, a continuous composite chain **102** is obtained which comprises an array of pairs of U-bent elastic thread portions **28**. The pairs of elastic thread portions **28** are mutually spaced in a longitudinal direction, and are joined together by a continuous tape **104** fixed to the ends **28′** of the elastic thread portions **28**.
(41) The continuous composite chain **102** is detached from wheel **76** in a detachment area. After releasing the elastic thread portions **28** in the detachment area, the movable sectors **92** move again into a position aligned with the fixed sectors **90**, in order to pick up further elastic thread portions in the entry section of wheel **76**.
(42) The ends **28′** of the elastic thread portions **28** may be fixed to the continuous tape by gluing, as in the presently described embodiment. In possible embodiments, the ends **28′** of the elastic thread portions **28** may be fixed to tape **104** by ultra-sonic welding or thermal welding. In this latter case, the glue dispensers **86**, **88** are omitted, whereas a welding unit is provided which performs welding between the ends **28′** of the elastic thread portions **28** and the non-woven layers **82**, **84**. In possible embodiments, the ends **28′** of the elastic thread portions **28** may be fixed to the non-woven layers **82**, **84** via mechanical anchoring spots, i.e. welding spots between the non-woven layers **82**, **84**, which are located on opposite sides of the ends **28′** of the elastic thread portions **28**, and which are spaced by a distance shorter than the diameter of the elastic threads **28** in an undeformed condition, as described in EP-B1-0886480 by Cera France.
(43) The continuous tape **104** may include two non-woven webs **82**, **84** overlapping each other, as in the embodiment shown in FIGS. **4** and **5**. In a possible embodiment, the continuous tape **104** may consist in a single web of non-woven fabric. The continuous tape **104** may be provided with printings arranged in phase relationship with the elastic thread portions **28**.
(44) In possible embodiments, instead of feeding continuous non-woven webs **82**, **84**, the apparatus may include cutting devices, configured to feed successions of discrete non-woven fabric portions mutually spaced in the longitudinal direction, and arranged at the ends **28′** of the U-bent elastic thread portions **28**.
(45) With reference to FIG. **4**, downstream wheel **76** there provided a is longitudinal cutting unit, which longitudinally cuts the continuous tape **104**. Directly downstream the longitudinal cutting unit **106** there are obtained a first and a second continuous array **108′**, **108″**. Each continuous array **108′**, **108″** includes a succession of ear loops **14**, each comprising a U-shaped elastic thread portion **28** and a tape portion **30** fixed to the opposite ends **28′** of the U-shaped elastic thread portion **28**. In each continuous array **108′**, **108″**, the tape portions **30** of mutually adjacent ear loops **14** are joined together by scrap tape portions **110**.
(46) The first and the second continuous arrays **108′**, **108″** are mutually spaced in a transverse direction, so that the distance between the ear loops **14** of each pair corresponds to the distance at

which the pairs of ear loops **14** must be applied to the respective mask bodies **12**.

(47) When longitudinal cutting is performed, the U-shaped elastic thread portions **28** are mutually aligned and facing outwardly of tape **104**. This arrangement is suitable for the case in which the ear loops **14** are applied to the mask bodies **12** with the elastic threads **28** extending outwardly of the respective mask bodies **12**.

(48) If it is desired to produce face masks **10** wherein the elastic threads **28** of the ear loops **14** are arranged within the mask bodies **12**, as in the embodiment of FIG. **1**, downstream the longitudinal cutting unit **106** both continuous arrays **108′**, **108″** may be rotated by 180° around the respective longitudinal axes, in a rotating device denoted as **112** in FIG. **3**. Directly downstream rotating device **112**, the elastic threads **28** are located within the two mutually parallel tapes.

(49) With reference to FIG. **6**, in order to produce masks **10** wherein the elastic threads **28** of the ear loops **14** are arranged within the mask bodies **12**, downstream the longitudinal cutting unit **106** both continuous arrays **108′**, **108″** may be picked up by respective conveyor belts **124**, **126** which cross each other and invert the relative position of the two continuous arrays **108′**, **108″**. The conveyor belts **124**, **126** may retain both continuous arrays **108′**, **108″** by suction. When they leave the conveyor belts **124**, **126**, the elastic threads **28** are arranged within the two mutually parallel tapes. This embodiment is advantageous because it enables a better control of both continuous arrays **108′**, **108″**, because both continuous arrays **108′**, **108″** are never detached and are not subjected to rotations as is the case in the previously described embodiment.

(50) Referring to FIG. **3**, apparatus **40** includes a cutting unit **114** configured to perform cutting of the tapes of both continuous arrays **108′**, **108″**. As shown in FIG. **4**, after this cutting individual ear loops **14** are obtained which respectively comprise a U-shaped elastic thread portion **28** and a tape portion **30** which is fixed to the opposite ends **28′** of the U-shaped elastic thread portion **28**. In the cutting step the tape portions **30** may be shaped in various ways.

(51) The individual ear loops **14** are arranged pairwise, the ear loops **14** of each pair being mutually aligned and spaced in a transverse direction (FIG. **4**). The ear loops **14** of each pair are mutually spaced in a transverse direction by a distance corresponding to the distance at which they will be applied to respective mask bodies **12**.

(52) In the cutting unit **114**, the scrap tape portions **110** positioned between the ear loops **14** of each continuous array **108′**, **108″** are detached from the ear loops **14** and disposed of as scrap, e.g. via a suction pipe.

(53) With reference to FIG. **3**, apparatus **40** comprises an application unit **116**, wherein the ear loops **14** are applied and attached to respective mask bodies **12**.

(54) The application unit **116** comprises an anvil roller **118** cooperating with a welding device **120**, e.g. an ultra-sonic welding device. The anvil roller **118** receives, from the transfer roller **67**, the mask bodies **12** oriented transversally with respect to the machine direction. The anvil roller **118** moreover receives an array of ear loops **14** coming from the cutting unit **114**. On the anvil roller **118**, the pairs of ear loops **14** are in-phase with the respective mask bodies **12**. The transfer roller **67** may be replaced by a repitch device, if the mask bodies **12** on the rotating device **66** have a different pitch with respect to the pitch between the successive ear loop pairs **14**.

(55) The welding device **120** cooperates with the anvil roller **118** in order to weld the tapes **30** of the ear loops **14** to the respective mask bodies **12**. In possible embodiments, the attachment of the tapes **30** of the ear loops **14** to the respective mask bodies **12** may be obtained by gluing or thermal welding.

(56) Downstream welding device **120** a continuous array of finished face masks **10** are obtained, which are transferred by the anvil roller **118** to an output conveyor **122**. Directly downstream welding device **120** there may be provided an inspection system, in order to visually inspect the finished face masks and check the correct configuration thereof, especially as regards the position and the presence of the elastic ear loops. This optical inspection may be performed with backlighting, in order to obtain a correct contrast.

(57) In a possible embodiment, the scrap tape portions **110** may be cut and removed after fixing the ear loops to the respective mask bodies **12**. In this case, the cutting unit **114** is arranged downstream welding device **120**.
(58) In a possible embodiment, the U-bent elastic thread portions **28** may be attached directly to the mask bodies **12**, without being previously fixed to a tape. In this case, apparatus **10** does not include the unwinders **78**, **80** for feeding the first and the second non-woven web **82**, **84**, and wheel **76** directly transfers the U-bent elastic thread portions **28** to the anvil roller **118**.
(59) As stated in the foregoing, the ear loops **14** are fixed to the respective mask bodies **12** with the U-shaped elastic threads **28** extending inwardly with respect to the mask bodies **12**. In a possible embodiment, the face masks **10** may be produced with the elastic threads **28** of the ear loops **14** oriented outwardly of the respective mask body **12**.
(60) The description in the foregoing refers to the production of surgical face masks wherein the mask body **12** consists in a rectangular panel provided with longitudinal pleats.
(61) In a possible embodiment, the method and the apparatus according to the present invention may be used for producing FFP filtering masks. In this case, the ear loop forming unit **68** is the same as previously described, whereas the mask body forming unit **42** is replaced with a unit configured to produce mask bodies of the FFP type. Generally speaking, the ear loops **14** produced by the ear loop forming unit **68** may be attached to various different mask bodies, produced by any conventional apparatus.
(62) The apparatus according to the present invention enables performing a continuous cycle process, with a production speed of about 600-1000 masks/1′, while keeping an accurate control of the length of the elastic threads forming the ear loops.
(63) The method and the apparatus according to the present invention enable the production of finished face masks without any direct human contact with the products. It is therefore possible to obtain a sterile packaging of the face masks, and to ensure the absence of contaminations.
(64) Of course, without prejudice to the principle of the invention, the implementation details and the embodiments may vary appreciably from what has been described and illustrated herein, without departing from the scope of the invention as defined by the Claims that follow.

Claims

1. An apparatus for manufacturing face protection masks having mask bodies, comprising: an ear loop forming unit configured to form a first and a second array of ear loops, each including a U-shaped elastic thread, an application unit configured to apply and attach pairs of ear loops to respective mask bodies, wherein said ear loop forming unit comprises a movable structure carrying two arrays of holding sectors, and wherein said holding sectors are arranged in groups of sectors configured to hold respective elastic threads portions, and a cutting roller configured to cut successive portions of elastic threads between each pair of groups of sectors adjacent to each other to form respective discrete elastic thread portions, and wherein each of said groups of sectors comprises a fixed sector and two movable sectors disposed laterally on opposite sides of the fixed sector and rotatable with respect to the fixed sector to U-bend the respective elastic discrete thread portions.
2. The apparatus according to claim 1, wherein said arrays of retaining sectors are carried by a wheel rotatable around its own axis.
3. The apparatus according to claim 1, wherein the ear loop forming unit comprises at least one unwinder configured to feed to said movable structure at least one non-woven web forming a tape to which ends of said discrete elastic thread portions are fixed.
4. The apparatus according to claim 3, wherein the ear loop forming unit comprises a longitudinal cutting unit configured to cut said tape in a longitudinal direction.